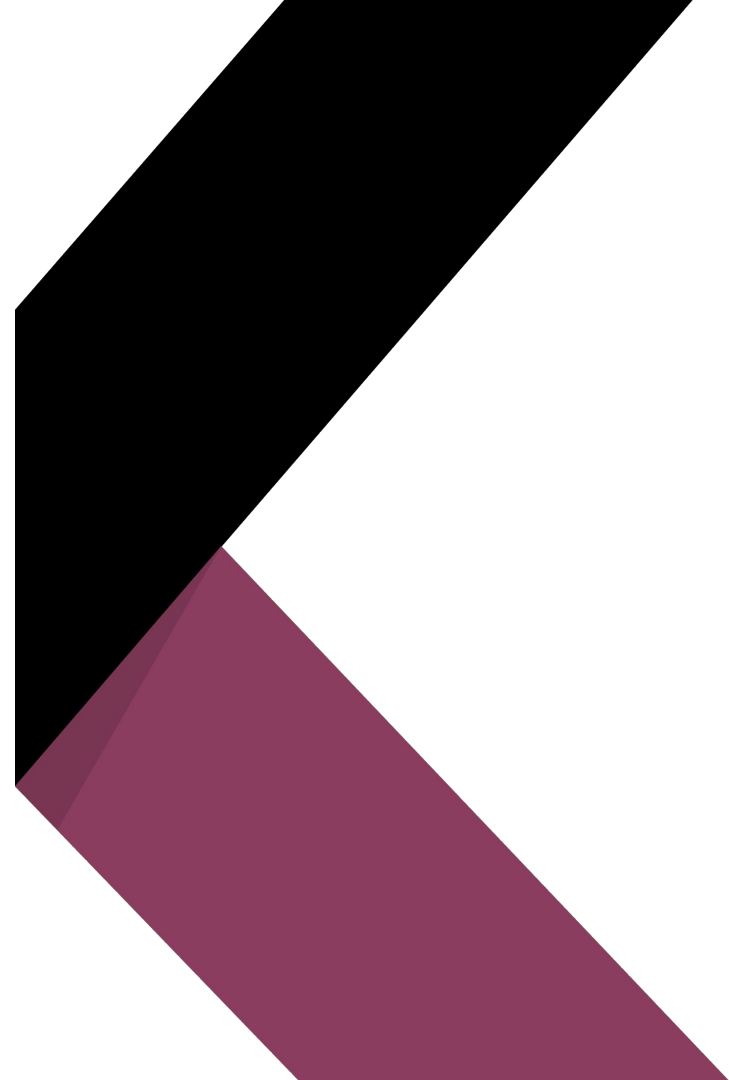


Diagnosis - Analysis

DIAnalyzer





DIAnalyzer



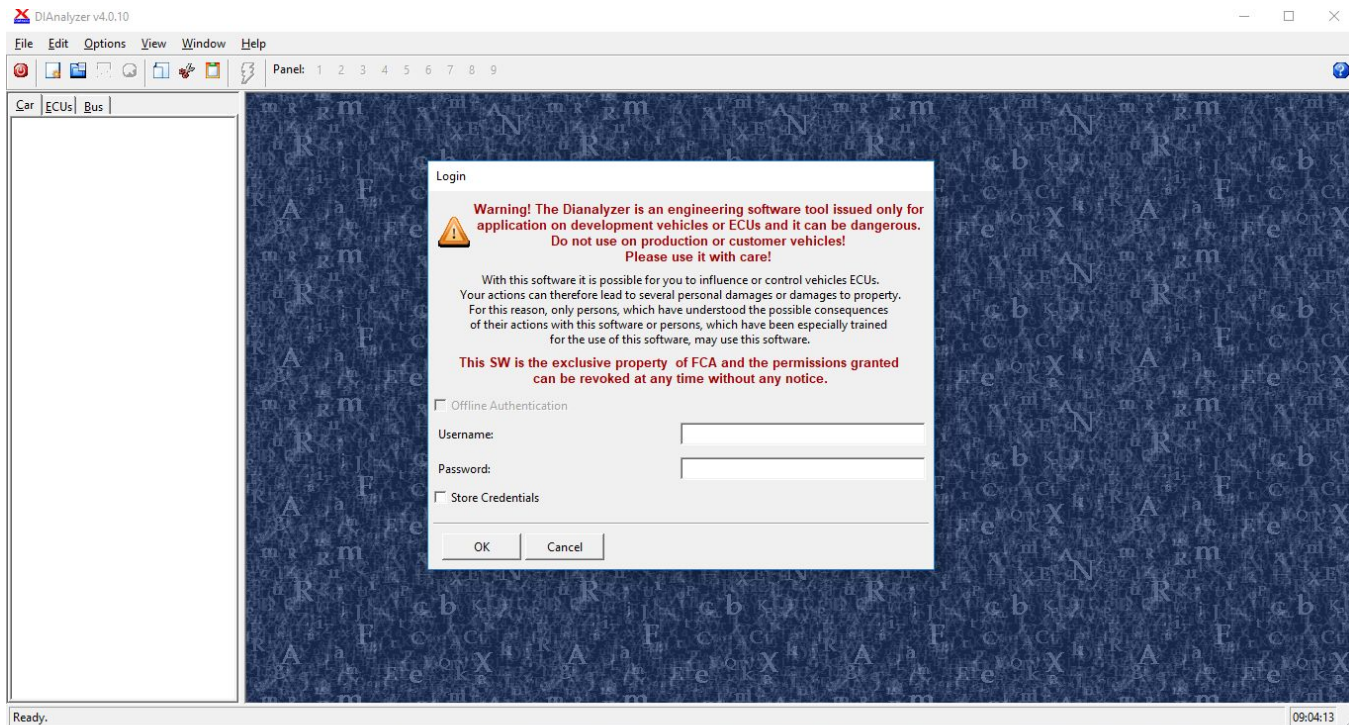
Dianalyzer - Introduction

Dianalyzer is the diagnostic tool that use FCA.

At the moment the actual version is 4.x.

This version is quite similar to CDA that is the diagnostic tool of Chrysler. It's necessary the login to use it (it's necessary the authorization).

It's possible to use it in Offline mode («Offline Authentication») but it's necessary a specific authorization.



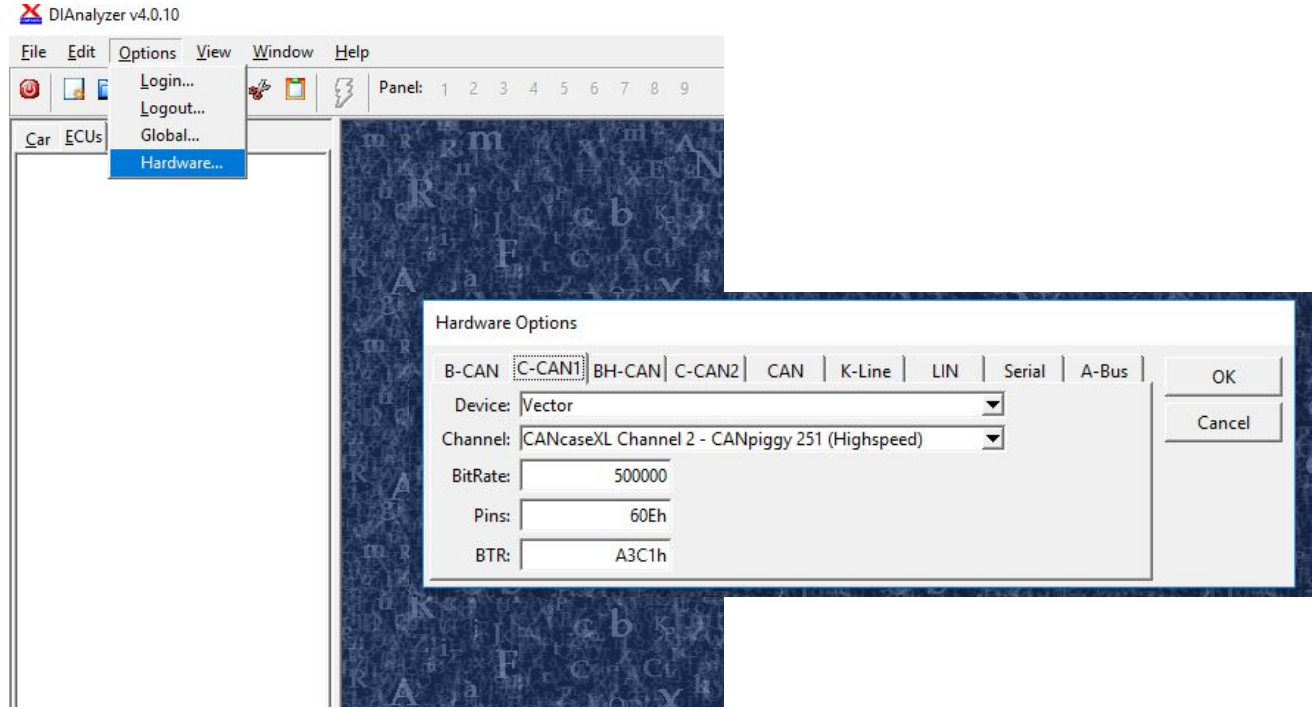


Hardware Selection

Options sheet □ Hardware: it's possible to explore the hardware setting windows for each CAN bus.

The selection of hardware and channel of the hardware is the first step to configure the correct communication between ECUs and DIAnalyzer.

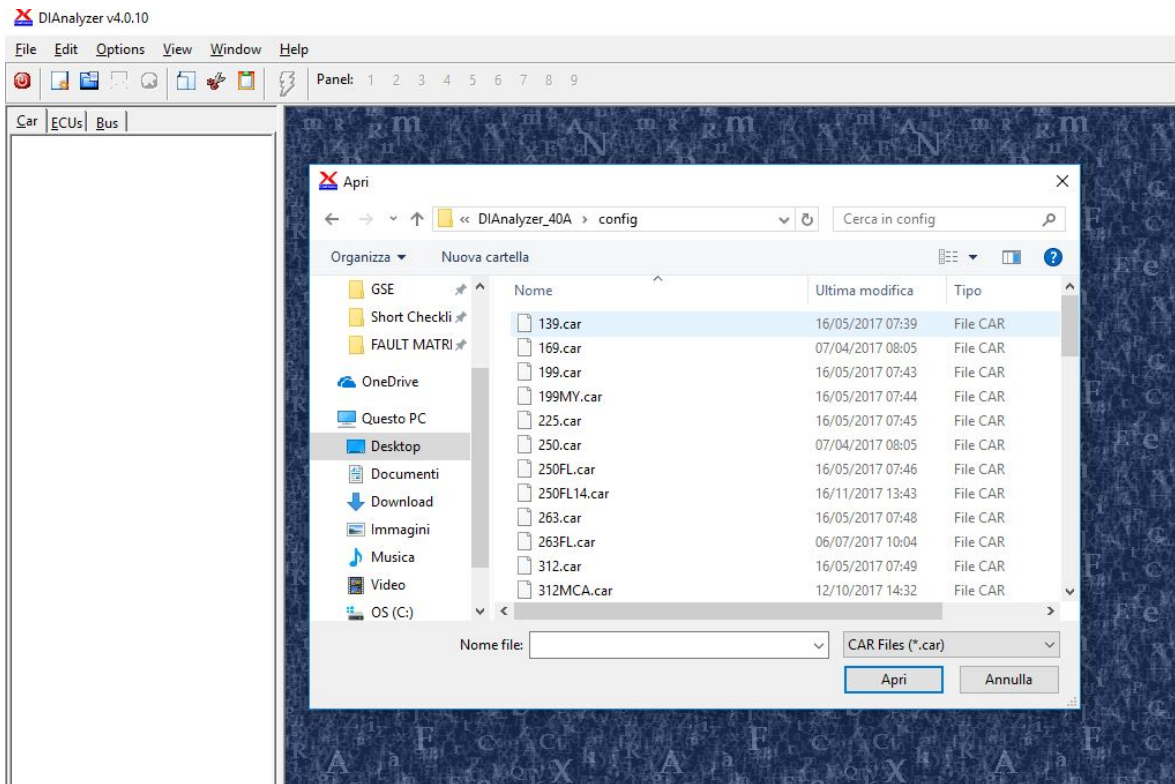
Note: the settings about the communication are depending from the hardware and from the used CAN bus □ for example, in this case, the ECM communicates with the Vector CancaseXL on channel 2 Highspeed.





The user has to select the .car of the project that is going to check:

File ☐ Open ☐ (select .car in the folder Config of DIAnalyzer)





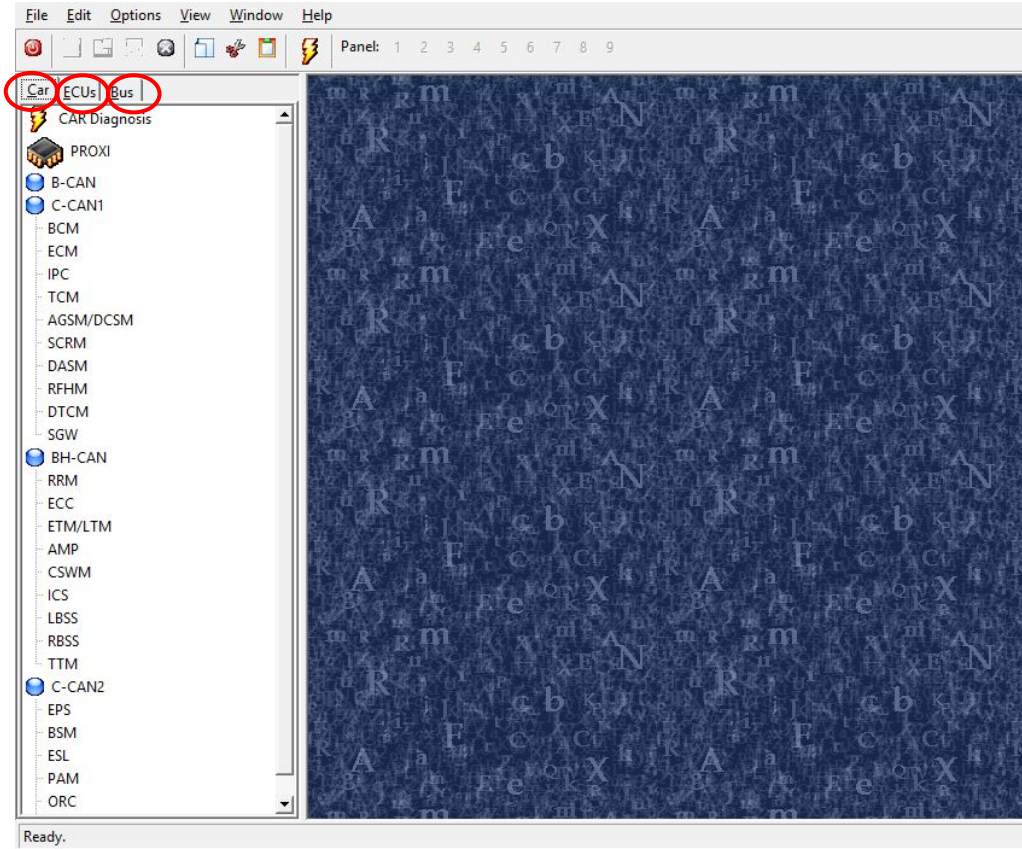
Main Screen

After the selection of the .car, DIAnalyzer fill the three sheets signed in **red** for the specific project:

Car ☐ in this section is shown all ECU of the vehicle in reference to the selected CAN bus.

ECUs ☐ in this section it's possible view all the vehicle's ECU (like as CDA)

Bus ☐ DIAnalyzer shows all the accessible bus of the vehicle.





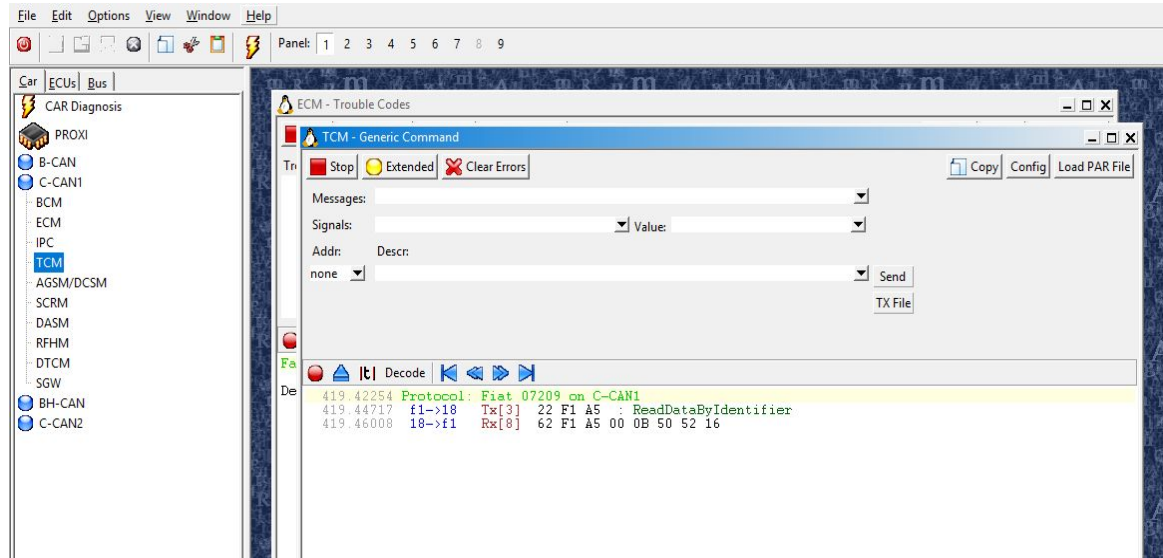
ECU Main Screen

It's possible to start the diagnostic communication between DIAnalyzer and a specific ECU clicking on the ECU for the selected bus ☐ the image below shows the connection with the TCM (Transmission Control Module, this is the ECU that it's necessary for each vehicle equipped with a non manual transmission).

If all hardware connection are correct, the communication starts. In the figure it's possible to check the Dianalyzer request (**Tx messages**) and the ECU response (messages **Rx**). In this case the response is positive.

On the left side of each frame it's possible to check:

- Communication timing for each frame
- ECU that are communicating (who is the sending? Who is the receiver)

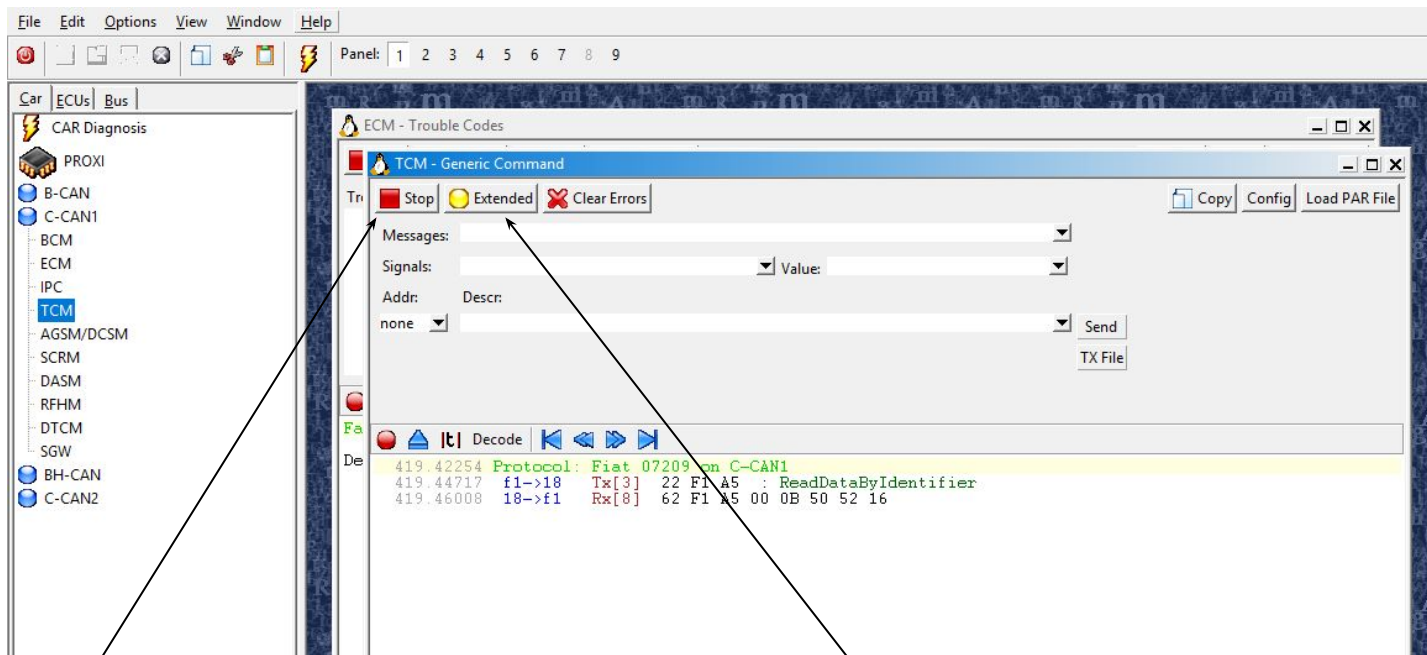




ECU - Generic Command

When the user selects the .car file, it's start the diagnostic communication with the ECU. DIAnalyzer upload in automatic way the .PAR file that it's necessary to decode the raw value exchanged between ECU and DIAnalyzer.

«Load PAR File» button needs to change the decryption file (it's necessary to decode the diagnostic information for each ECU).



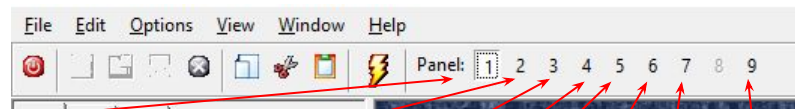
«Stop» button needs to stop the communication between DIAnalyzer and the selected ECU. If the user presses the button, this button change the name in «Start»; in that case, the selection of this button needs to restart the communication with the ECU.

«Extended» button needs to switch between the many diagnostic sessions that are necessary to interact with the ECU: **Default** / **Extended**.



Overview of all

Screen



Screen 1 – Generic Command

Generic screen, it's possible read the communication «trace» and it's possible to send any kind of request to the ECU in raw format (in this screen it's possible read the raw response of the ECU).

Screen 2 – ECU Identifications

In this screen DIAnalyzer shows the ECU identification parameters, like SW number, HW number, ISO code, etc...

Screen 3 – Diagnostic Trouble Codes

Here, DIAnalyzer shows all the stored DTC (Diagnostic trouble code); there are all the stored DTC in the diagnostic memory of the ECU; there are also the state and details of each error like snapshot and extended data record.

Screen 4 – Parameter Reading

DIAnalyzer shows all the RDBI (Read data by Identifier) necessary for diagnostic analysis. The RDBI are called also engineering parameters

Screen 5 – Active Diagnosis

DIAnalyzer can command many actuators to perform an active diagnosis with IOCBi function (Input Output Control by identifier).

Screen 6 – EOL Programming (PROXI)

In this panel it's possible to perform the ECU personalization procedure (PROXI).

Screen 7 – Cyber Security

Screen 9 – Download

In this panel it's possible to perform a software download on the ECU.



Screen 1 – Raw command

The image shows an user examplet for Raw command.

User requires to the ECU all the DTC stored (with a raw request) 19 02 FF.

ECU response (coherent and positive) 59 02 FF ...

The screenshot displays a diagnostic software interface with a menu bar (File, Edit, Options, View, Window, Help) and a toolbar. On the left, a tree view shows various vehicle components under 'CAR Diagnosis', including PROXI, B-CAN, C-CAN1, BCM, ECM, IPC, TCM, AGSM/DCSM, SCRM, DASM, RFHM, DTCM, SGW, BH-CAN, RRM, ECC, ETM/LTM, AMP, CSWM, ICS, LBSS, RBSS, TTM, C-CAN2, BSM, ESL, PAM, and ORC. The main window is titled 'ECM - Generic Command' and contains a 'Messages' dropdown set to 'ReadDiagnosticTroubleCode'. Below this, a 'Signals' dropdown is set to 'Value:'. The 'Addr' field shows 'Descr: 3 byte: 19 02 FF ...' and the 'Tx' field shows '19 02 FF'. A 'Send' button is visible. The bottom section shows a list of raw data packets with columns for time, address, and data. The data is as follows:

Time	Address	Data
60.01194	18->f1	Rx[13] 59 06 1C BE 00 0E 01 60 80 28 60 81 00
60.01202	f1->18	Tx[3] 19 02 08 : ReadDiagnosticTroubleCode
60.02542	18->f1	Rx[7] 59 02 FF 1C BE 00 0E
60.02552	f1->18	Tx[6] 19 04 1C BE 00 FF : ReadDiagnosticTroubleCode
60.07191	18->f1	Rx[57] 59 04 1C BE 00 0E 00 0B 10 08 00 00 DE AB 10 09 00 40 20 0A 1F 90 60 82 01
60.07203	f1->18	Tx[6] 19 06 1C BE 00 FF : ReadDiagnosticTroubleCode
60.08741	18->f1	Rx[13] 59 06 1C BE 00 0E 01 60 80 28 60 81 00
60.08747	f1->18	Rx[3] 19 02 08 : ReadDiagnosticTroubleCode
60.10194	18->f1	Rx[7] 59 02 FF 1C BE 00 0E
60.10288	f1->18	Tx[6] 19 04 1C BE 00 FF : ReadDiagnosticTroubleCode
60.14741	18->f1	Rx[57] 59 04 1C BE 00 0E 00 0B 10 08 00 00 DE AB 10 09 00 40 20 0A 1F 90 60 82 01
60.14748	f1->18	Tx[6] 19 06 1C BE 00 FF : ReadDiagnosticTroubleCode
60.16341	18->f1	Rx[13] 59 06 1C BE 00 0E 01 60 80 28 60 81 00
60.16349	f1->18	Tx[3] 19 02 08 : ReadDiagnosticTroubleCode
60.17742	18->f1	Rx[7] 59 02 FF 1C BE 00 0E
60.17752	f1->18	Tx[6] 19 04 1C BE 00 FF : ReadDiagnosticTroubleCode
60.22342	18->f1	Rx[57] 59 04 1C BE 00 0E 00 0B 10 08 00 00 DE AB 10 09 00 40 20 0A 1F 90 60 82 01
60.22353	f1->18	Tx[6] 19 06 1C BE 00 FF : ReadDiagnosticTroubleCode
60.24069	18->f1	Rx[13] 59 06 1C BE 00 0E 01 60 80 28 60 81 00
71.55294	f1->18	Tx[3] 19 02 FF : ReadDiagnosticTroubleCode
71.75110	18->f1	Rx[299] 59 02 FF 07 16 00 40 07 16 00 40 07 16 00 40 07 16 00 40 27 66 00 40 27 6



Screen 2 – ID

DIAnalyzer requires and shows all ECU ID like Software number, Hardware number, ECU serial number, Spare Part number, etc...

The screenshot displays the DIAnalyzer software interface. On the left, a tree view under 'Car' shows various ECUs, with 'C-CAN2' selected. The main window is titled 'ECM - Generic Command' and contains a sub-window 'TCM - Identification'. This window has buttons for 'Stop', 'Extended', 'Pause', and 'Clear Errors', along with 'Copy', 'Config', and 'Load PAR File'. It includes checkboxes for 'Full Data' and 'Loop'. Below these, a hex dump shows data from address 34 to 43. The data is decoded into several fields:

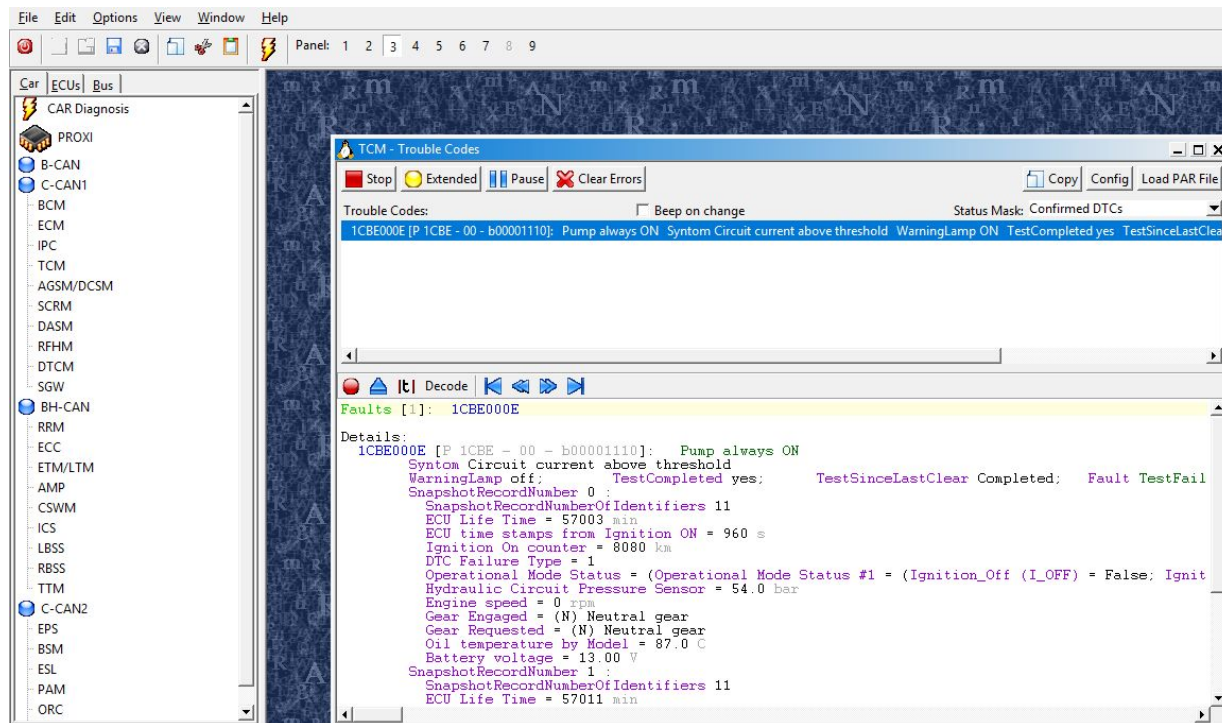
- Identifications:** SparePartNumber (46336021), VIN Current (????????????????), VIN Original (????????????????), HW Number (CFC 8TDW.01), HW Version (00), SW Number (UN16BB26), SW Version (00 00), Omolog. Code (), Sincom&Factory (), ISO Code (00 0B 50 52 16), ECUSerialNumber (41 31 52 30 30 53 41 31 52 20 20 20 20 20 20).
- SoftwareIdentification:** Boot (NumberOfModules = 1, ModuleID = 01, Version = 34 39 5F 36 31 20 20 20 0D 03, Supplier), Application (NumberOfModules = 1, ModuleID = 01, Version = 55 4E 31 36 20 20 20 20 20 20, Supplier), Data (NumberOfModules = 1, ModuleID = 01, Version = 42 42 32 36 20 20 20 20 20 20, Supplier).
- Fingerprint:** Boot (ModuleID = 20, Tester = , ProgDate = 20/20/20), Application (ModuleID = 20, Tester = , ProgDate = 20/20/20), Data (ModuleID = 20, Tester = , ProgDate = FF/20/20).
- Vehicle Manufacturer:** ECU Software Number (52104254), ECU Hardware Number (52028977).



Screen 3 – DTC management

Read DTC from memory ☐ if the user selects the DTC, in the lower section of the screen it appears the details of environment condition during DTC setting (☐ snapshot and extended data record).

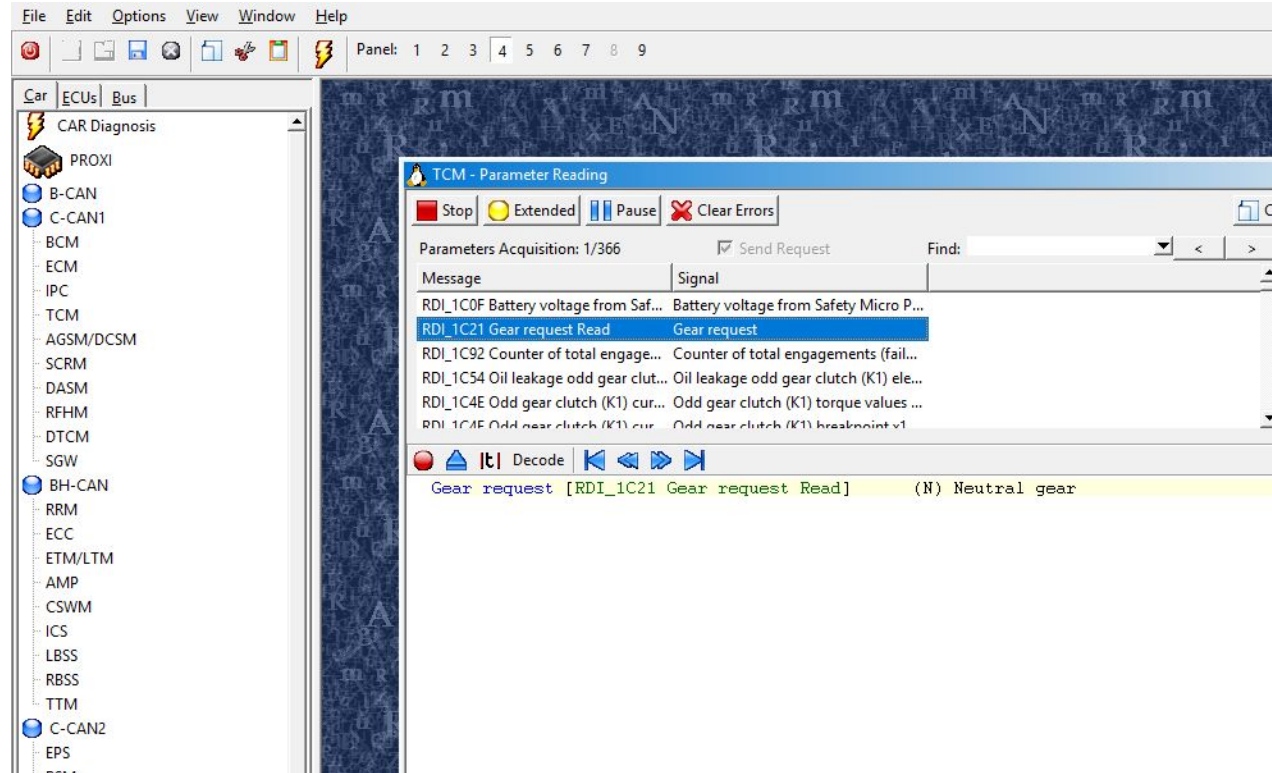
Note: it's important that the user choose the correct .par because it's the only way to decode correctly all the necessary information (DTC description, RDI decoding, ecc...)





Screen 4 – RDI management

When the user selects a RDI, DIAnalyzer shows the ECU response (the response is already decoded). The decode of data is correct only if the user choose the correct .par file.





Screen 5 – Active Diagnosis (IOCBI)

DIAnalyzer can execute an Active diagnosis only in Extended Session

The user has to select the actuator ID, the type of command (e.g. shortTermAdjustment) and the value of the request (FF = On, 00 = Off).

The screenshot shows the DIAnalyzer software interface. The left sidebar lists various ECUs, with 'C-CAN2' selected. The main window displays a list of active diagnosis codes (52AF to 52B5) and their descriptions. The '52B5 Shift Lock command Control' is highlighted. On the right, a control panel shows buttons for 'Stop', 'Default', 'Pause', and 'Clear Errors', along with a 'shortTermAdjustment ON/OFF' dropdown menu set to 'Off'. The bottom section shows a list of CAN bus messages with their IDs, addresses, and data fields.

code	I/O	Description
52AF	Engagement electrovalve PPV2 co...	
52B1	Range selection electrovalve comm...	
52B2	Odd gear clutch (K1) control solen...	
52B3	Even gear clutch (K2) control solen...	
52B4	Electric pump drive unit command ...	
52B5	Shift Lock command Control	

Active Diagnosis:

returnControlToECU (0x00)

1179.25999 18->f1 Rx[6] 50 03 00 32 01 F4 : ExtendedDiagnostic Start :
P2Can = 50
P2CanExtended = 5000
1181.72821 f1->18 Tx[4] 2F 52 B5 00 : InputOutputControlByIdentifier
1181.73553 18->f1 Rx[3] 7F 2F 22 : ConditionsNotCorrect
1184.30974 f1->18 Tx[4] 2F 52 B5 01 : InputOutputControlByIdentifier
1184.31584 18->f1 Rx[3] 7F 2F 22 : ConditionsNotCorrect
1187.17418 f1->18 Tx[4] 2F 52 B5 02 : InputOutputControlByIdentifier
1187.17962 18->f1 Rx[3] 7F 2F 31 : RequestOutOfRange
1190.07665 f1->18 Tx[5] 2F 52 B5 03 FF : InputOutputControlByIdentifier
1190.08343 18->f1 Rx[4] 6F 52 B5 03
1194.08404 f1->18 Tx[2] 3E 00 : TesterPresent
1194.09147 18->f1 Rx[2] 7E 00 : Tester Present Tester Ppresent
1198.09944 f1->18 Tx[2] 3E 00 : TesterPresent
1198.10006 18->f1 Rx[2] 7E 00 : Tester Present Tester Ppresent
1199.57031 f1->18 Tx[5] 2F 52 B5 03 00 : InputOutputControlByIdentifier
1199.57556 18->f1 Rx[4] 6F 52 B5 03
1203.57530 f1->18 Tx[2] 3E 00 : TesterPresent
1203.58335 18->f1 Rx[2] 7E 00 : Tester Present Tester Ppresent
1207.58396 f1->18 Tx[2] 3E 00 : TesterPresent
1207.59116 18->f1 Rx[2] 7E 00 : Tester Present Tester Ppresent



Screen 9 – ECU Software Download

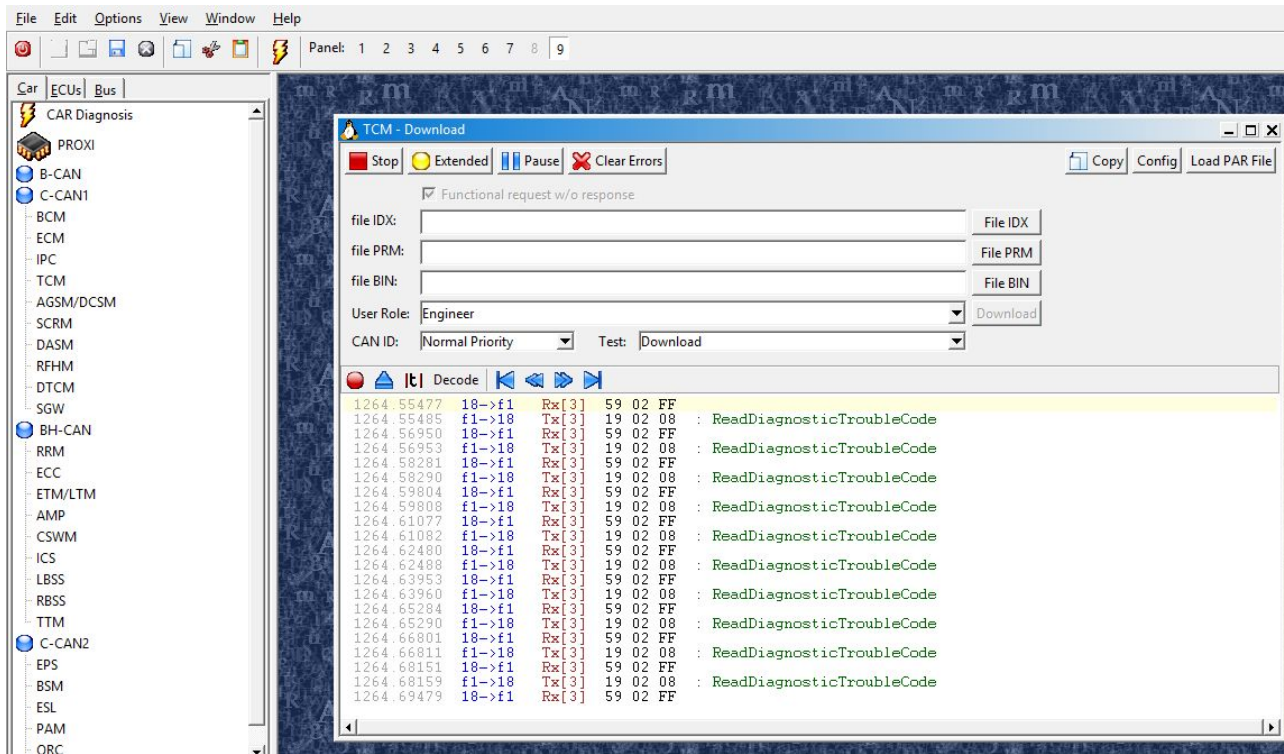
To update an ECU with DIAnalyzer it's necessary the presence of three different file: .idx, .prm e .bin.

The user has to put these three file (with the same name but with different extension) in the same folder.

The procedure for the software download is:

- Press «File IDX»: select the file located on the PC;
- DIAnalyzer select in an automatic way the other two files.
- If DIAnalyzer can't select the .bin and the .prm, the user can perform this operation in manual mode;
- The user has to press the Download button;
- DIAnalyzer shows the progress of the download procedure and it shows a popup message when the procedure has terminated.

Note: it's possible to open the .idx file with a text editor to check the content.





Screen 6 – PROXI procedure 1\3

PROXI procedure.

FCA requires the development of a software general purpose to his software supplier. FCA requires to develop a software that can be customizable (for different engine, transmission and optional). FCA requires a software that can manage or not any kind of functionality; in that way, the supplier can develop only one software and FCA can personalize this software in the production site (PROXI procedure).

The PROXI procedure is a configuration process: with the PROXI the use can activate/deactivate many functionalities during the EOL (End Of Line) process; for example the use can personalize the vehicle with/without Stop&Start, with manual/automatic climate control, gear ratio and other functionalities.

Screen PROXI – it's recommended to access using the icon PROXI in the CAR screen.

The use can perform the PROXI procedure in KeyOn (it's not possible perform the procedure in Engine Running).

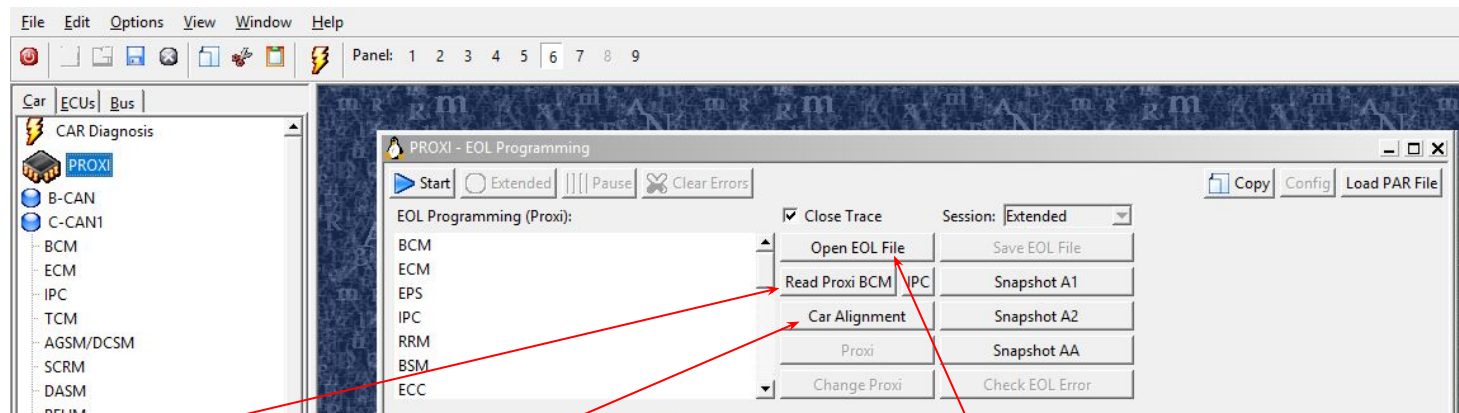
Only the qualified users can perform the PROXI procedure.

Each vehicle has his configuration file: in the configuration file there are all the features of the vehicle.

The configuration file is stored in the BCM and a backup copy is located in the IPC



Screen 6 – PROXI procedure 2\3



Read Proxi BCM:

With this command the user can read the configuration file of the vehicle and he can save this file with the extension .byt.

This functionality is possible only in the vehicle because it's necessary the presence of a real BCM. In any case it's possible also read the configuration file from the IPC

Car Alignment:

The user can command the BCM to force the PROXI on all vehicle ECU: in the specific, the BCM provides the personalization command (they are present in the configuration file) to all ECU of the vehicle.

Open EOL File:

This functionality is useful to work in a simulated environment (e.g. HiL simulator or stati bench). If an user saved a configuration file from a real BCM, it's possible to save it with extension .byt and it's possible open and modify.



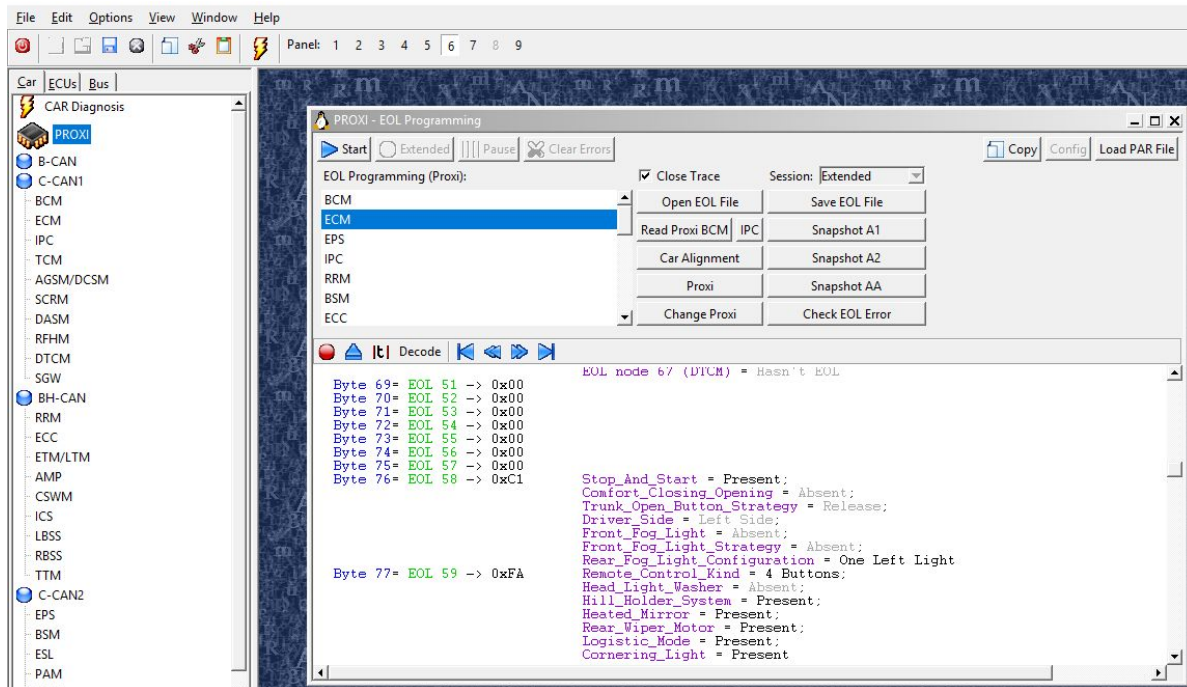
Screen 6 – PROXI procedure – Example

It's necessary the next step:

- Upload of a .PAR file for the specific project ☐
- Upload of a .byt file with the button «Open EOL File».

In the example there's the results of the operation described above:

- In the lower section of the screen there're the decoding of the .byt file content; here there are all the activable features of the ECU
- It's possible to modify the activable features of the ECU using the button «Change Proxi»
- It's useful save each time the .byt file with a different name; you can distinguish between the original configuration and the new ones.



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